

ME-203A: Kinematics of Machines

by

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SYLLABUS:

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

UNIT 2

Special purpose mechanism: Mechanisms for straight line, Motorcar steering mechanisms, Hooks joint and Geneva mechanism

UNIT 3

Kinematics synthesis of mechanism: Type of synthesis, DOF, Grashof's criterion of movability, Limit position, Dead center transmission angle synthesis of mechanism by relative pole method, Inversion method, Freudenstein's methods of synthesis, Design of a four bar chain for constant angular velocity ratio of cranks.

UNIT 4

Friction: Solid, rolling greasy and viscous friction, Laws of Friction inclined plane, Friction of pivots and collars, Friction circle of turning pair and friction axis of a link, Ball and roller bearings, Single and multiple disc-clutches

UNIT 5

Belts, Ropes & Chains: Mechanics of belt drive, Length of belt, Ratio of tensions in flat and V belt, Effect of centrifugal force, Power transmission capacity, Belt materials.

Rope drive: Types of ropes, Construction, Forces on rope and breaking strength,

Chain drive: Types, Chain length, Chordal action hoisting and power transmission

ME 203 A: KINEMATICS OF MACHINES

2L, 1T

3 Hours, 100 Marks

COURSE OUTCOMES:

CO1: Describe the concepts of mechanisms, machines and related terminologies and discuss different types of methods to analyse the velocity and acceleration of various mechanism.

CO2: Discuss Geneva mechanism.

CO3: Design of a four bar chain for constant angular velocity ratio of cranks.

CO4: Explain the concepts of friction and its types. In addition, assess its effects of different mechanical components.

CO5: Analyze various motion transmission elements like belt drive, rope drive and chain drive.

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

Reference Books:

1. Theory of Machines, Kinematics and Dynamics, By Sadhu Singh, Published by Pearson.
2. Theory of Machines, By Thomas Bevan, Published by Pearson.
3. Theory of Machines, By S. S. Rattan, Published by Mc Graw Hill Education.
4. Theory of Machine, R.S. Khurmi and J.K. Gupta, Published by Eurasia Publishing House (Pvt.) Ltd.

Definition of Kinematics:

- ❖ **Study of motion** of particles and rigid bodies without reference to forces.

Mechanisms and Machines:

- ❖ **Kinematics of inter-connected rigid bodies.**

Equilibrating forces:

- ❖ **quasi-static analysis.**

Mechanism:

Combination of rigid bodies (links) so shaped and connected that they move relative to each other for the purpose of transferring and transforming motion / forces from input (s) (actuator (s)) to an output.

Machine:

A device that uses mechanism (s) to transform and transfer motion and force from a power source to perform some useful mechanical work.

Mechanisms and Machines:

- ❖ Transmission and transformation of motion and forces.

Motion:

- ❖ Displacement, velocity, acceleration, path.

Forces transmission: actuator forces

Motivation:

- **Mechanization and Automation:** requirement of specialized mechanical devices (manufacturing, mechanical handling, assembly, painting, packaging).
- **Health-care:** transfer aids and devices, physiotherapy surgery.
- **Synthesis and design:** requires kinematic analysis.

Link or Element:

- ❖ Each part of a machine which has motion relative to some other part is termed as element or link.
- ❖ A link need not necessarily be a rigid body, but it must be a resistant body. It must be capable of transmitting the required force with negligible deformation.

Examples: links which are resistant, but not rigid.

1. Liquids, which are resistant to compressive forces and are used as links in hydraulic presses, i.e. hydraulic brakes and hydraulic jacks.
2. Chains, belts and ropes; which are resistant to tensile forces and are used for transmitting motion and forces.

Classification of Links:

1. Binary link
2. Ternary link
3. Quaternary link

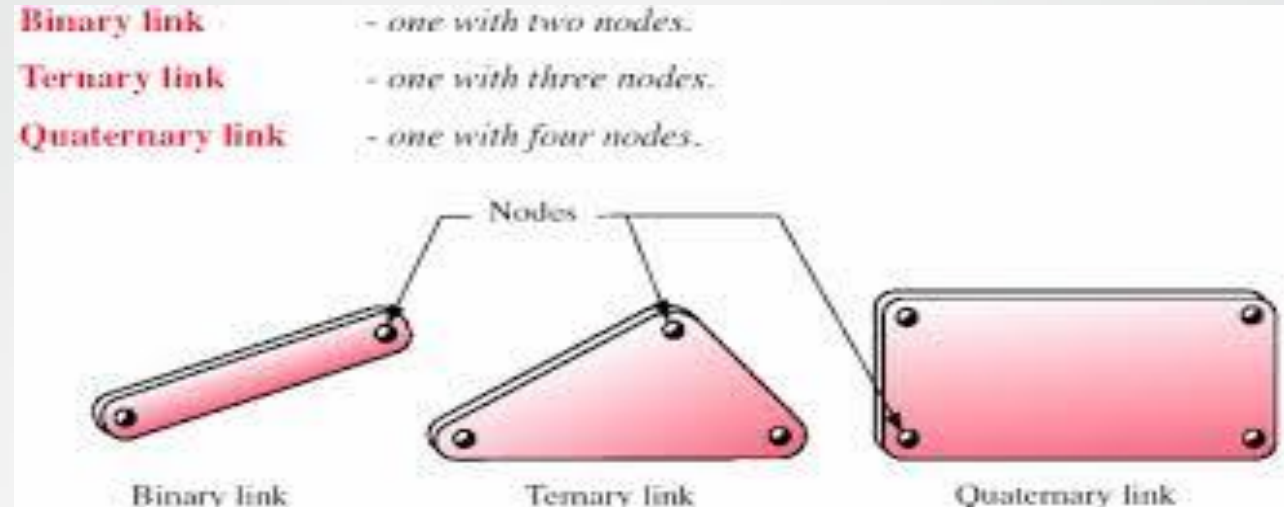


Fig. Reference mechediploma.com

There are four types of links:

1. **Rigid link**: a rigid link does not undergo any deformation while transmitting motion. Links in general are elastic in nature. E.g. connecting rod, crank, etc.
2. **Flexible link**: A flexible link is one which while transmitting motion is partly deformed in a manner not to effect the transmission of motion. E.g. belts, ropes, chains, springs, etc.
3. **Fluid link**: A fluid link is deformed by having fluid in a closed vessel and the motion is transmitted through the fluid by pressure. E.g. hydraulic press, hydraulic jack and fluid brake.
4. **Floating link**: It is a link which is not connected to the frame.

Kinematic Pair:

- ❖ The two links of a machine, when in contact with each other, are said to form a pair.
- ❖ Two elements or links which are connected together in such a way that their relative motion is completely constrained from a kinematic pair.

Classification of Kinematic Pairs:

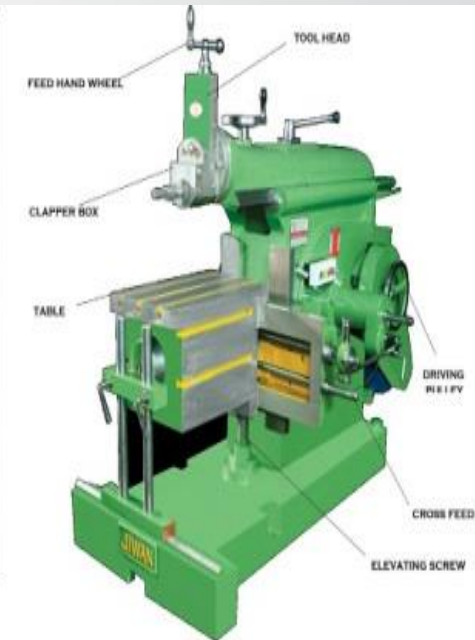
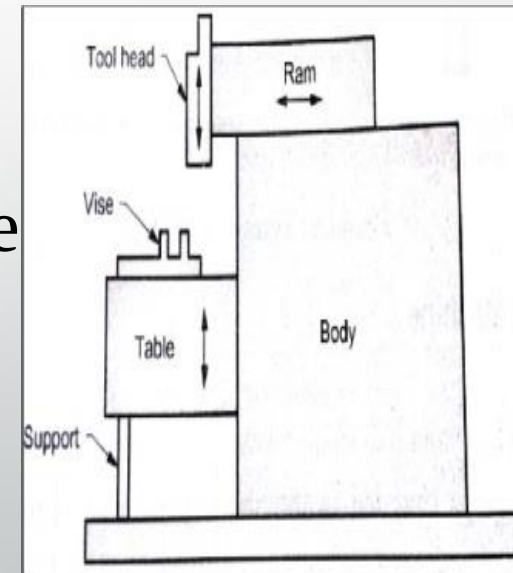
1. Type of relative motion
2. Type of contact
3. Type of mechanical constraint

1. Kinematic pairs according to the relative motion:

- **Sliding pair:** It consists of two elements connected in such a manner that one is constrained to have sliding motion to another.

For example:

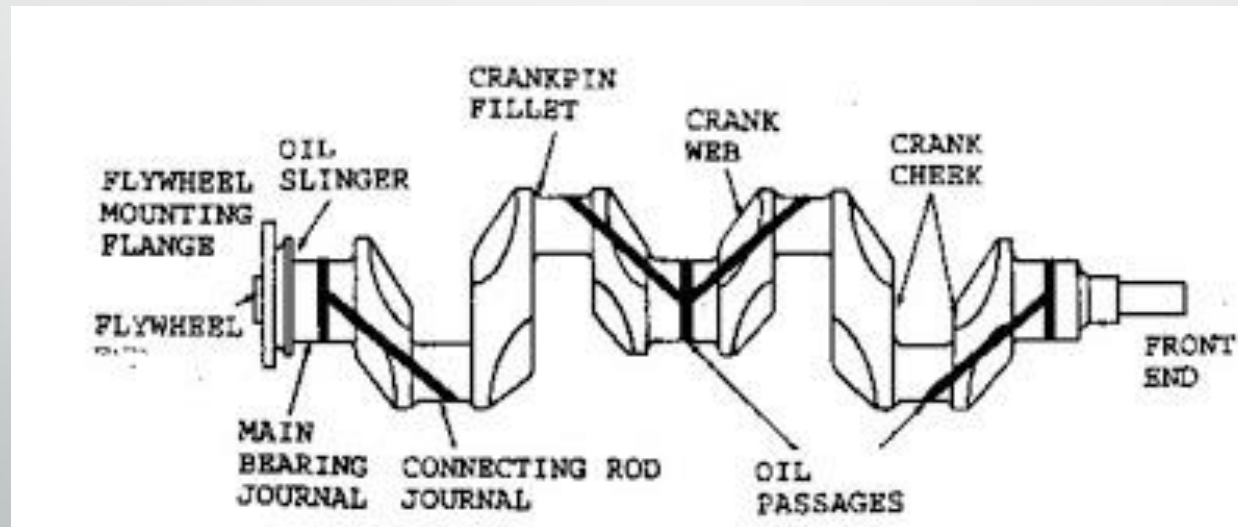
1. rectangular bar in rectangular hole
2. Piston and cylinder of an engine
3. Cross-head and guides of a steam engine
4. Ram and its guides in a shaper
5. Tailstock on the lathe bed



- **Turning (revolute) pair:** It consists of two elements connected in such a manner that one is constrained to turn or revolve about a fixed axis of another element.

For example:

1. a shaft with collar at both ends revolving in a circular hole
2. Crankshaft turning in a bearing
3. Cycle wheels revolving over their axles.



- **Roller pair:** when two element are connected that one is constrained to roll on another element which is fixed, form a rolling pair.

For example:

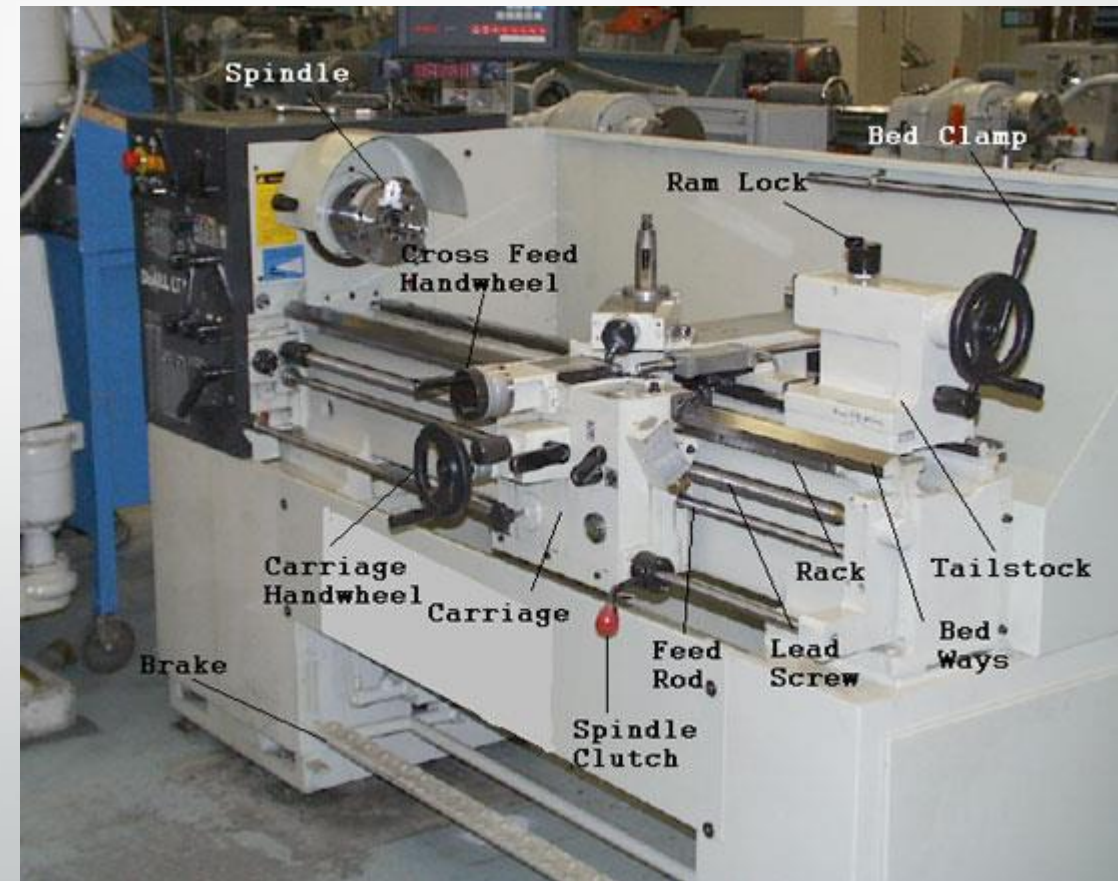
1. Ball and roller bearing
2. A wheel rolling on a flat surface



- **Screw (or helical) pair:** when one element turns about the other element by means of threads, it forms a pair. The motion in this case is a combination of sliding and turning.

For example:

1. Lead screw of a lathe machine with nut
2. Screw with nut of a jack
3. Bolt with nut



- **Spherical pair:** when one element in the form of a sphere turns about the other element, it forms a spherical pair.

For example:

1. Ball and socket joint
2. Pen stand
3. The mirror attachment of vehicles

Ball & Socket Joint
eg. Hip Joint

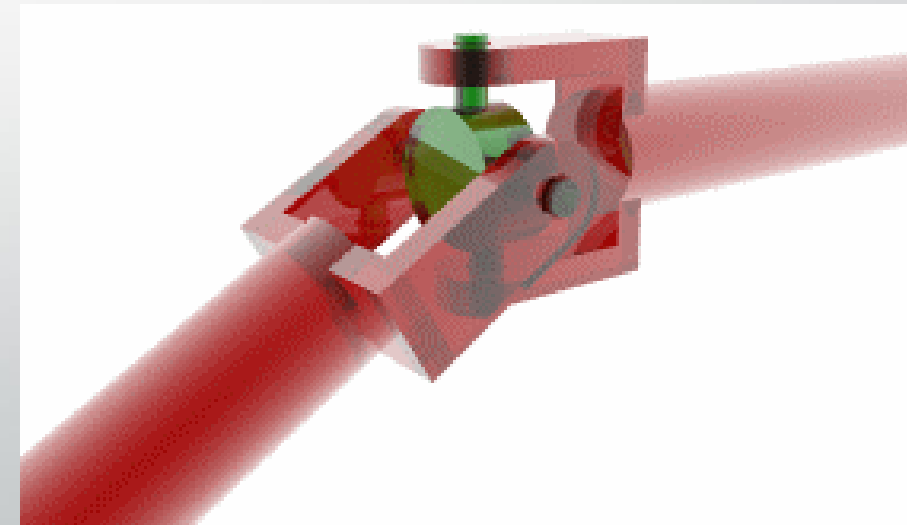


2. Kinematic pairs according to the type of contact:

- **Lower pair:** when two elements have surface (area) contact while in motion and relative motion is purely turning or sliding, they are called a lower pair.
- All the sliding pair, turning pair, and screw pair form a lower pair.

For example:

1. Nut turning on a screw
2. Shaft rotating in a bearing
3. Universal joint
4. All pairs of slider crank mechanism
5. Pantograph

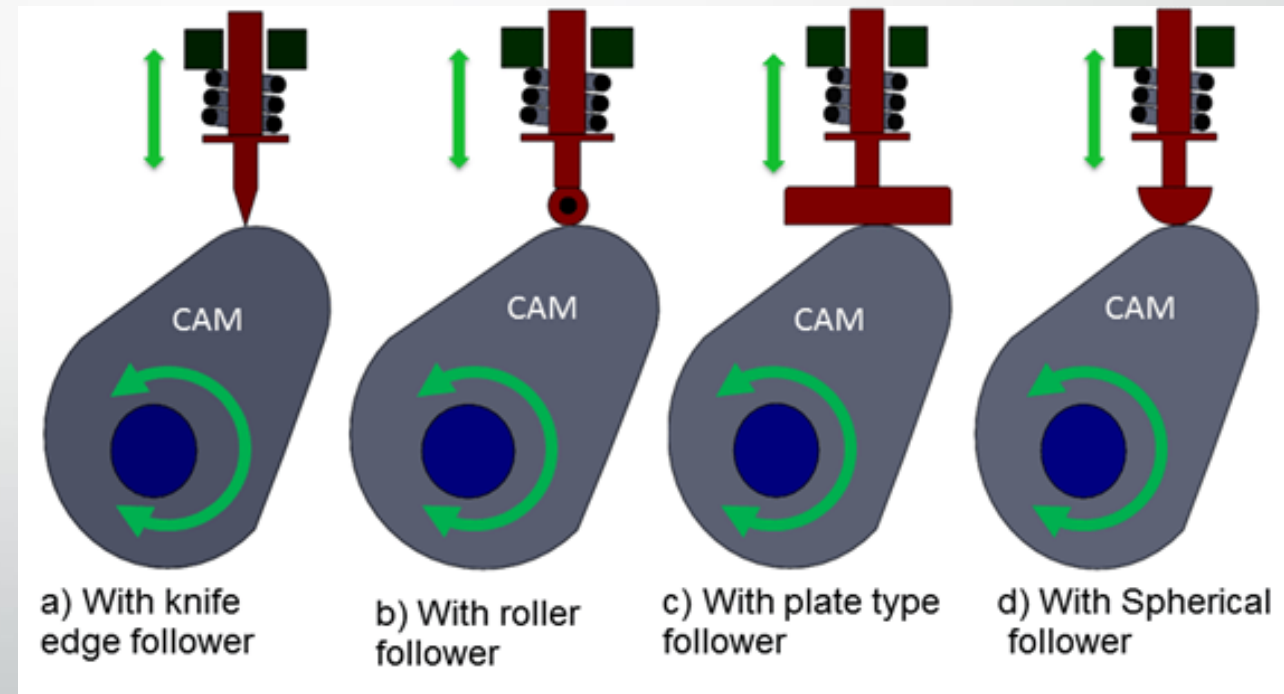


Reference: youtube.com

- **Higher pair:** when two elements have point or line contact while in motion and relative motion is purely turning or sliding, then the pair is known as higher pair.

For example:

1. Belts
2. Ropes
3. Chain drive
4. Gears
5. Cam and follower
6. Ball and roller bearings
7. Wheel rolling on a surface



3. Kinematic pairs according to Mechanical constraint:

- **Closed pair:** when two elements of a pair held together mechanically in such a manner that only the required type of relative motion occurs, they are called closed pair.

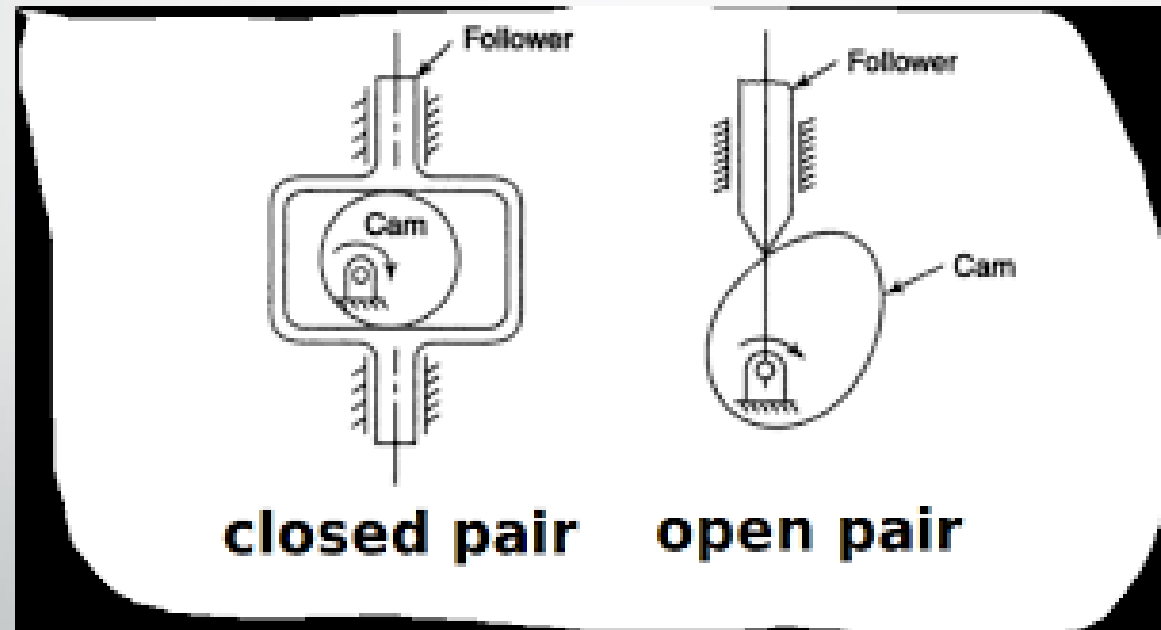
For example:

1. All the lower pairs and some of higher pair (enclosed cam and follower) are closed pairs.

- **Unclosed pair:** when two elements of a pair are not held mechanically and are held in contact by the action of external forces, are called unclosed pair.

For example:

1. Cam and spring loaded follower pair

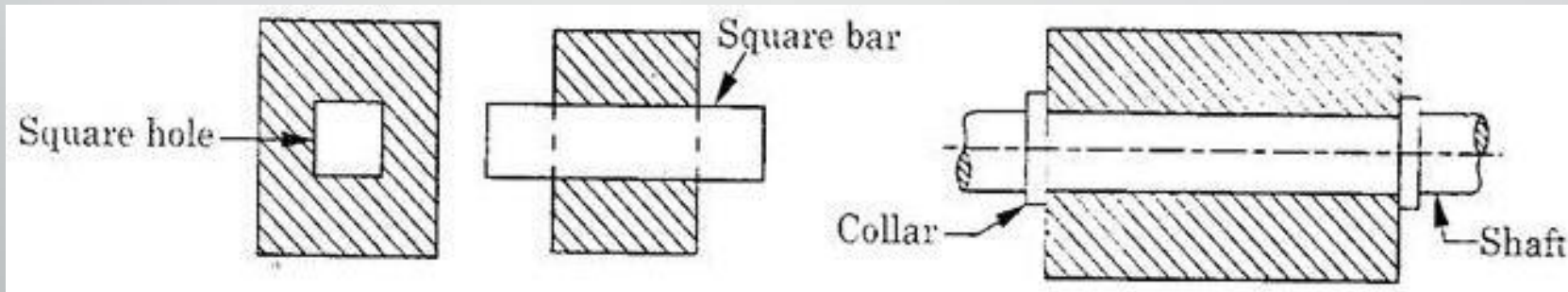


Constrained Motion

- 1. Completely constraint motion:** when the motion between a pair takes place in a definite direction irrespective of the direction of force applied, then the motion is said to be a completely constrained motion.

For Example:

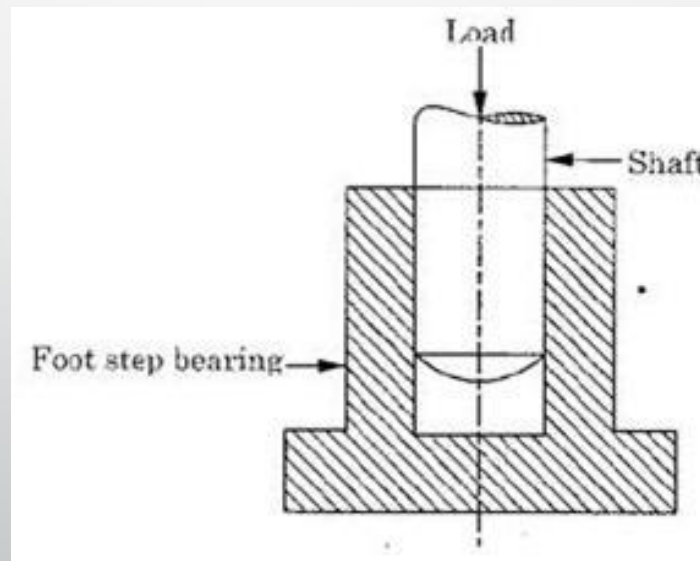
1. A square bar in square hole
2. A shaft with collars at each end in a circular hole
3. A piston in the cylinder of an internal combustion engine



2. Partially or successfully constrained motion: when the constrained motion between a pair is not completed itself but by some other means, then the motion is said to be a successfully constrained motion.

For Example:

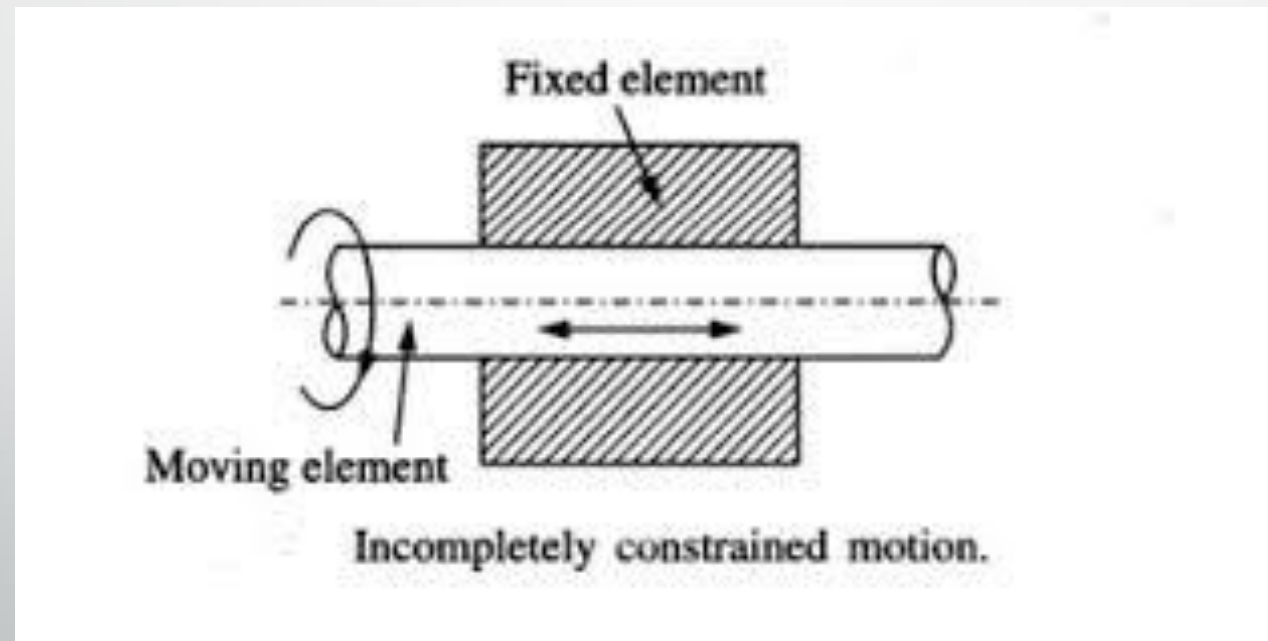
1. The motion of a shaft in a footstep bearing becomes successfully constrained motion when compressive load is applied to the shaft.



3. Incompletely constrained motion: when the motion between a pair can take place in more than one direction, it is said to be incompletely constrained motion.

For Example:

1. A circular shaft in a circular hole.





Thank You